Lab 4 - Lunar Lander

AI-assisted work record · PES2UG24CS593 · 15 October 2026

Record type: Retrospective summary of the work actually performed in this chat. It is not a verbatim or reconstructed prompt-by-prompt transcript. The linked ChatGPT page contains the real conversation.

1. Assignment and repository

The allocation workbook maps SRN PES2UG24CS593 to repository 32_lunarLander. The repository README specifies a touchdown horizontal-speed fix plus three features: fuel-sensitive ship color, a successful-landing effect, and an extra-life score threshold.

2. Implementation

README item	Change made
Landing validation	Added abs(horizontal speed) <= MAX_SPEED_X to the safe touchdown conditions.
Fuel color	Interpolate the hull from pale blue at full fuel to red at empty fuel; clamp the fuel fraction to [0, 1].
Landing feedback	Create a score-colored particle burst at the landing position and draw it briefly after touchdown.
Bonus life	Return a 1,500-point threshold; existing score bookkeeping awards one life per newly crossed threshold.

3. Checks and evidence

Python syntax compilation succeeded. A behavior check confirmed that high sideways speed crashes, a safe landing succeeds, the fuel-color endpoints are correct, and crossing 1,500 points awards one life. Both MP4 files were confirmed as 10 seconds, 30 fps, 800 × 600.

Video method: The before and after clips are scripted demos rendered from the Pygame drawing code. They stage the target outcomes; they are not unedited screen recordings of a person playing.

4. Current repository status

A local Git repository was initialized on branch main. The Lab-4 files are staged, but there is no initial commit because no commit email is configured. No remote was set and nothing was pushed. To finish the assignment upload, the local repo still needs the student's personal Lab repo remote and a commit identity.

ChatGPT history link supplied by student

Open the supplied ChatGPT history link

https://chatgpt.com/share/6ad11b67-2bac-83ec-aa74-ea2aff62043c

Submission must follow the course instructions: place these deliverables under the personal repository's Lab-4 folder and do not open a pull request against SETAPESU26.